Document Intelligence Refiner

Interim Report

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1. Domain Notes

During the development of the Document Intelligence Refinery pipeline, I analyzed the four documents provided in the challenge dataset located in data/raw/data. These documents represent several common document classes encountered in financial and administrative archives.

The dataset includes the following documents:

1. 2013-E.C-Assigned-regular-budget-and-expense.pdf
2. 2013-E.C-Audit-finding-information.pdf
3. 2013-E.C-Procurement-information.pdf
4. 2018_Audited_Financial_Statement_Report.pdf

Although these documents belong to similar administrative domains, they exhibit different structural properties that influence the extraction strategy required.

Document Class 1 — Native Digital Financial Report

Example document

2018_Audited_Financial_Statement_Report.pdf

This document contains a digital text layer and structured formatting typical of financial reports generated using document authoring software.

Observed properties:

- machine-readable text layer present
- consistent page layout
- moderate text density

Failure risk:

- tables may appear flattened when extracted with simple text tools
- layout ordering may be lost

Extraction strategy used:

Strategy A — Fast Text Extraction

Document Class 2 — Scanned Administrative Records

Example documents:

2013-E.C-Assigned-regular-budget-and-expense.pdf

2013-E.C-Audit-finding-information.pdf

These documents appear to be scanned copies of printed administrative records.

Observed properties:

- no embedded text layer
- page images occupy most of the page area
- text must be reconstructed using OCR

Failure mode:

Direct text extraction returns empty results.

Extraction strategy used:

Strategy B — Layout-aware OCR extraction

Document Class 3 — Mixed Layout Procurement Documents

Example document:

2013-E.C-Procurement-information.pdf

This document contains complex layout structures including tabular procurement records.

Observed properties:

- mixed layout containing text blocks and tables
- irregular spacing between elements
- partial page scanning artifacts

Failure mode:

Simple OCR may read rows incorrectly if layout structure is not preserved.

Extraction strategy used:

Strategy B — Layout-aware extraction using Docling

Document Class 4 — Complex Documents (Potential Strategy C Cases)

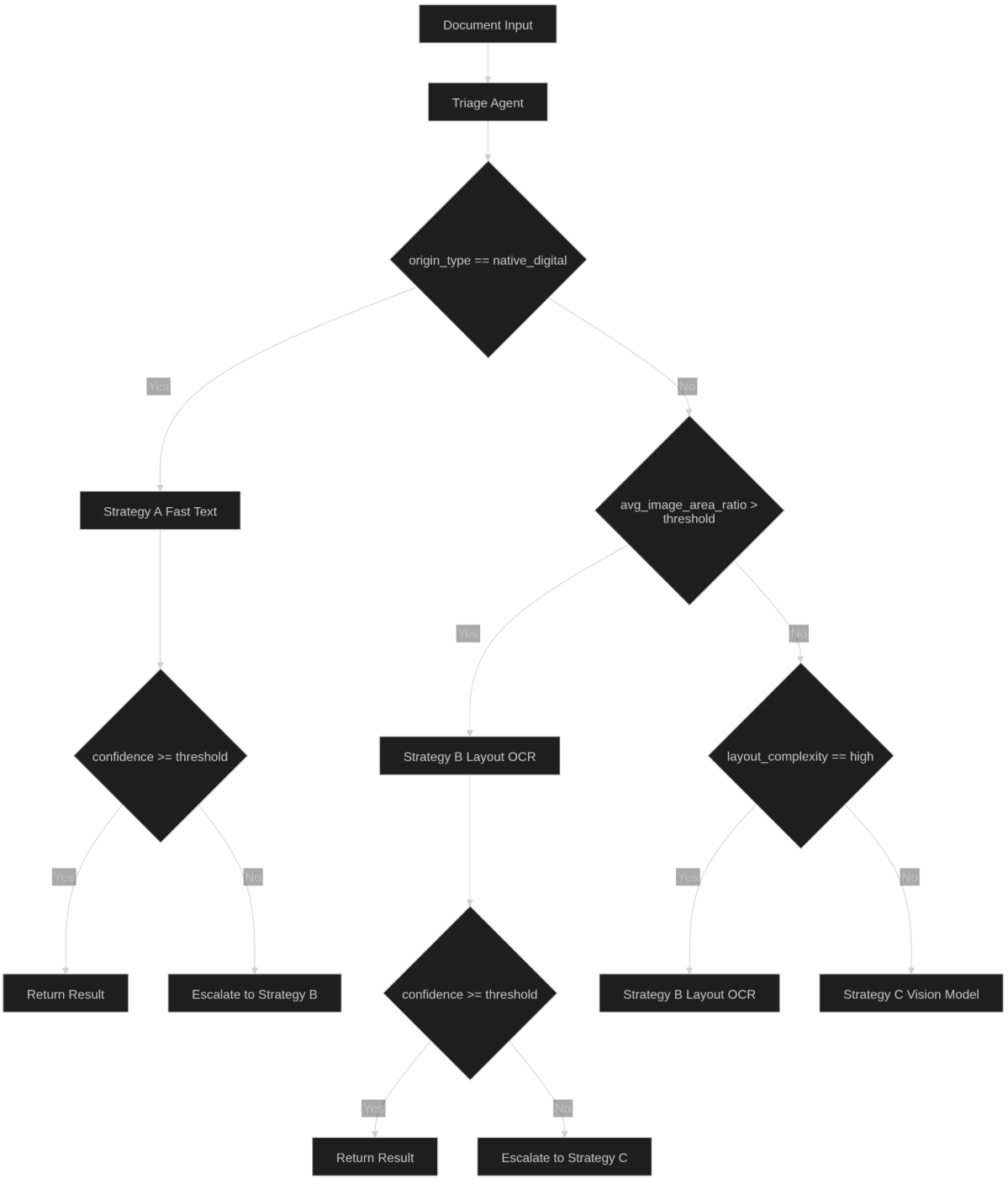
While the current corpus did not strictly require Strategy C, certain documents show characteristics that could require a vision-language model in larger datasets.

Example indicators include:

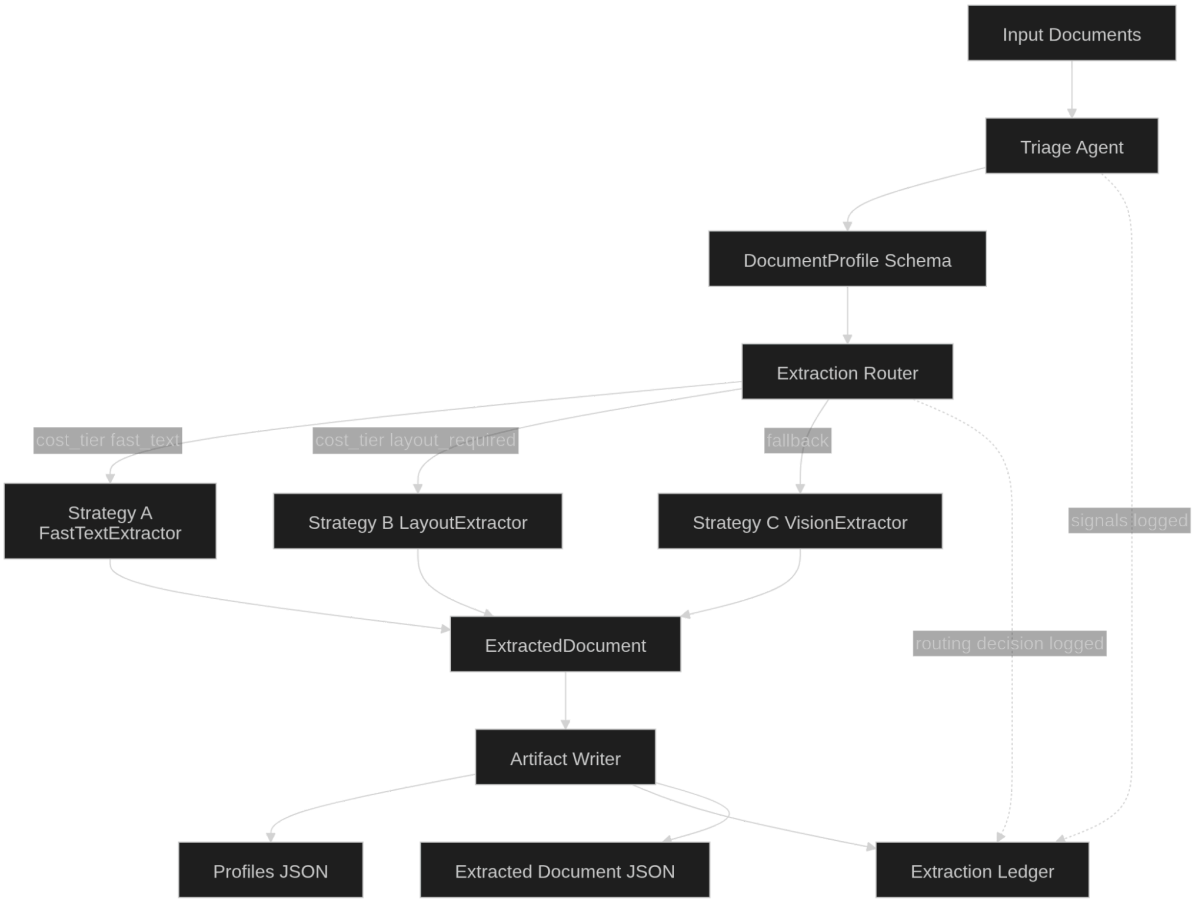
- heavy graphical layout
- embedded charts
- mixed visual elements

Strategy C is therefore designed as a fallback for such document**s.**

Strategy Decision Tree



2. ARCHITECTURE DIAGRAM



Cost Analysis

The system is designed to minimize processing cost by selecting the least expensive extraction strategy capable of producing acceptable results.

The cost estimates are derived from three components:

1. computational complexity of the extraction algorithm
2. average processing time per page
3. expected infrastructure cost for OCR or model inference

Strategy A — Fast Text Extraction

Tools used:

pdfplumber
PyMuPDF

Processing characteristics:

- direct text extraction from PDF structure
- no OCR required

Estimated processing time:

~0.005 seconds per page

Estimated cost:

$0.001 per document

Quality tradeoff:

This strategy works well for documents containing a digital text layer but may lose table structure.

Strategy B — Layout-Aware OCR Extraction

Tools used:

Docling
RapidOCR

Processing characteristics:

- image rendering
- layout detection
- OCR inference

Estimated processing time:

~2–3 seconds per page

Estimated cost:

$0.01 per document

Quality improvement:

This strategy successfully extracts text from scanned documents and preserves layout structure.

Strategy C — Vision-Language Model Extraction

Tools considered:

Vision-Language Models

Processing characteristics:

- image encoding
- transformer inference

Estimated processing time:

10–20 seconds per document

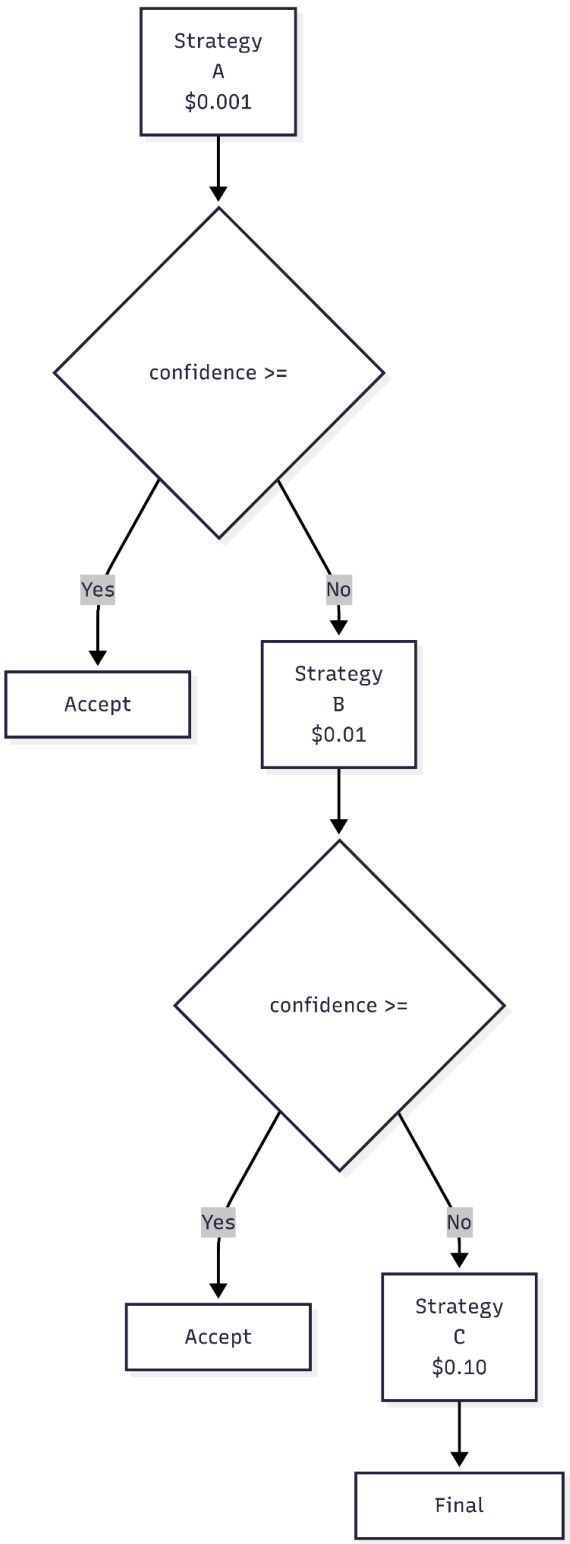
Estimated cost:

$0.10 per document

Quality improvement:

Strategy C can interpret complex documents containing diagrams, tables, and mixed visual elements.

Final Cost Escalation Diagram


Strategy
A
$0.001
confidence >=
Yes
No
Accept
Strategy
B
$0.01
confidence >=
Yes
No
Accept
Strategy
C
$0.10
Final